

LM7805 3-Terminal 5V Regulator

1. Device Overview & Absolute Maximum Ratings

The LM7805 is a 3-terminal monolithic positive voltage regulator designed for local, on-card regulation. Features internal current limiting, thermal shutdown protection, and safe operating area protection.

Parameter	Symbol / Rating	Recommended Operating Value
Input Voltage Range	VI = 7.0V to 35.0V	Optimal Vin = 7.5V to 12.0V
Output Voltage	VO = 5.0V (+/-4%)	4.8V min, 5.2V max at 1A
Maximum Output Current	IO = 1.5A (with heatsink)	0.5A continuous without heatsink
Operating Temperature	TJ = 0°C to 125°C	Thermal shutdown at TJ > 150°C
Dropout Voltage	VD = 2.0V typical	Vin must exceed Vout by >= 2.0V

Note: If input voltage drops below 7.0V, the device falls out of regulation and output drops below 5.0V.

2. Pinout Configuration (TO-220 & TO-92 Packages)

Looking at the front of the TO-220 package (metal tab facing backward):

- Pin 1 (INPUT): Unregulated DC input from power supply or battery (+7V to +35V DC).
- Pin 2 (GROUND): Common circuit ground. Connected internally to the metallic mounting tab/heatsink.
- Pin 3 (OUTPUT): Regulated +5.0V DC output rail.

Thermal Warning: Because Pin 2 is connected to the heatsink tab, connecting the tab to an external chassis will tie that chassis directly to electrical ground.

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3. Mandatory Bypass Capacitors & Typical Application Circuit

Bypass capacitors are required to suppress high-frequency oscillations and handle rapid load transients:

- Input Capacitor (Cin): 0.33 uF ceramic capacitor placed within 2 cm of Pin 1 to counteract input lead inductance.
- Output Capacitor (Cout): 0.1 uF ceramic capacitor on Pin 3 to improve transient response and phase margin.
- Bulk Filter Capacitor: When powered from an AC rectified supply, place a 1000 uF electrolytic capacitor upstream of Cin.

4. Troubleshooting & Thermal Failure Modes

- Problem 1: Regulator is boiling hot and output drops to ~1.2V. Cause: Thermal Overload Shutdown. Power dissipation $P = (V_{in} - V_{out}) * I_{load}$. For 12V input at 0.5A, $P = (12 - 5) * 0.5 = 3.5$ Watts. Without a heatsink, the junction temperature exceeds 150°C in seconds. Solution: Add extruded aluminum heatsink or lower input voltage to 7.5V.
- Problem 2: Output measures 5.0V on multimeter, but microcontroller resets randomly. Cause: High frequency oscillation. The multimeter averages the voltage, but Cin or Cout is missing, causing 200 mV oscillation spikes. Solution: Solder 0.1 uF ceramic capacitors directly adjacent to the regulator pins.
- Problem 3: Output voltage is 0V. Cause: Reverse polarity on Pin 1, short circuit between Pin 3 and Ground, or input voltage under 7.0V.